WIPRO TALENTNEXT PROGRAM

PBL HANDS-ON ASSIGNMENT

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Topic: Using Single-Row Functions to Customize The Output

Question 1

Write a query to display the current date. Label the column Date.

Query:

SELECT SYSDATE AS “Date” FROM DUAL;

Output:

```
SQL> SELECT SYSDATE  AS “Date” FROM DUAL;

Date
---------
09-JUL-26
```

Question 2

The HR department needs a report to display the employee number, last name, salary, and salary increased by 15.5% (expressed as a whole number) for each employee. Label the column New Salary.

Query:

SELECT EMPNO,ENAME,SAL, ROUND(SAL*1.155) AS “New Salary” FROM EMP;

Output:

```
Command Prompt - sqlplus /

SQL> SELECT SYSDATE  AS “Date” FROM DUAL;

Date
---------
09-JUL-26

SQL> SELECT EMPNO,ENAME,SAL ROUND(SAL*1.155) AS “New Salary” FROM EMP;
SELECT EMPNO,ENAME,SAL ROUND(SAL*1.155) AS "New Salary" FROM EMP
                             *
ERROR at line 1:
ORA-00923: FROM keyword not found where expected


SQL> SELECT EMPNO,ENAME,SAL, ROUND(SAL*1.155) AS “New Salary” FROM EMP;

     EMPNO ENAME             SAL New Salary
---------- ---------- ---------- ----------
      7000 DEEPU             800        924
      7001 ASHU             1600       1848
      7002 BHANU            1250       1444
      7003 JONES            2975       3436
      7004 MANI             1250       1444
      7698 BALU             2850       3292
      7788 SCOTT            3000       3465
      7844 TURNER           1500       1733
      7900 JAMES             950       1097
      7839 AISHU            5000       5775
      7006 SAM              8000       9240

11 rows selected.

SQL> |
```

Question 3

Modify the previous query to add a column alias that subtracts the old salary from the new salary. Label the column Increase.

Query:

SELECT EMPNO,ENAME,SAL,ROUND(SAL*1.155) AS “New Salary”,ROUND((SAL*1.155)-SAL) AS “Increase “ FROM EMP;

Output:

```
SQL> SELECT EMPNO,ENAME,SAL,ROUND(SAL*1.155) AS “New Salary”,ROUND((SAL*1.155)-SAL) AS “Increase “ FROM EMP;

     EMPNO ENAME             SAL New Salary  Increase
---------- ---------- ---------- ---------- ----------
      7000 DEEPU             800        924        124
      7001 ASHU             1600       1848        248
      7002 BHANU            1250       1444        194
      7003 JONES            2975       3436        461
      7004 MANI             1250       1444        194
      7698 BALU             2850       3292        442
      7788 SCOTT            3000       3465        465
      7844 TURNER           1500       1733        233
      7900 JAMES             950       1097        147
      7839 AISHU            5000       5775        775
      7006 SAM              8000       9240       1240

11 rows selected.
```

Question 4

Write a query that displays the ename (with the first letter uppercase and all other letters lowercase) and the length of the ename for all employees whose name starts with the letters J, A, or M. Give each column an appropriate label. Sort the results by the employees' enames.

Query:

SELECT INITCAP(ENAME) AS "Employee Name", LENGTH(ENAME) AS "Name Length" FROM EMP WHERE ENAME LIKE 'J%'   OR ENAME LIKE 'A%' OR ENAME LIKE 'M%' ORDER BY ENAME;

Output:

```
SQL> SELECT INITCAP(ENAME) AS "Employee Name",LENGTH(ENAME) AS "Name Length"FROM EMP WHERE ENAME LIKE 'J%'   OR ENAME LIKE 'A%' OR EN
AME LIKE 'M%' ORDER BY ENAME;

Employee N Name Length
---------- -----------
Aishu                5
Ashu                 4
James                5
Jones                5
Mani                 4
```

Question 5

Rewrite the query so that the user is prompted to enter a letter that starts the last name. For example, if the user enters H when prompted for a letter, then the output should show all employees whose last name starts with the letter H.

Query:

SELECT INITCAP(ENAME) AS "Employee Name", LENGTH(ENAME) AS "Name Length" FROM EMP WHERE ENAME LIKE '&LETTER%' ORDER BY ENAME;

Output:

```
SQL> SELECT INITCAP(ENAME) AS "Employee Name",  LENGTH(ENAME) AS "Name Length" FROM EMP WHERE ENAME LIKE '&LETTER%' ORDER BY ENAME;
Enter value for letter: M
old   1: SELECT INITCAP(ENAME) AS "Employee Name",  LENGTH(ENAME) AS "Name Length" FROM EMP WHERE ENAME LIKE '&LETTER%' ORDER BY ENAM
E
new   1: SELECT INITCAP(ENAME) AS "Employee Name",  LENGTH(ENAME) AS "Name Length" FROM EMP WHERE ENAME LIKE 'M%' ORDER BY ENAME

Employee N Name Length
---------- -----------
Mani                 4

SQL>
```

Question 6

The HR department wants to find the length of employment for each employee. For each employee, display the ename and calculate the number of months between today and the date on which the employee was hired. Label the column MONTHS_WORKED. Order your results by the number of months employed. Round the number of months up to the closest whole number.

Query:

SELECT ENAME, CEIL(MONTHS_BETWEEN(SYSDATE, HIREDATE)) AS MONTHS_WORKED FROM EMP ORDER BY MONTHS_WORKED;

Output:

```
SQL> SELECT ENAME, CEIL(MONTHS_BETWEEN(SYSDATE, HIREDATE)) AS MONTHS_WORKED FROM EMP ORDER BY MONTHS_WORKED;

ENAME      MONTHS_WORKED
---------- -------------
SCOTT                471
JAMES                536
MANI                 538
TURNER               539
BALU                 543
JONES                544
ASHU                 545
BHANU                545
DEEPU                547

9 rows selected.
```

Question 7

Create a report that produces the following for each employee: earns monthly but wants <3 times salary>. Label the column Dream Salaries**.**

Query:

SELECT ENAME || ' earns monthly ' || SAL || ' but wants ' || (SAL*3) AS "Dream Salaries"

FROM EMP;

Output:

```
SQL> SELECT ENAME || ' earns monthly ' || SAL ||   ' but wants ' || (SAL*3) AS "Dream Salaries" FROM EMP;

Dream Salaries
--------------------------------------------------------------------------------
DEEPU earns monthly 800 but wants 2400
ASHU earns monthly 1600 but wants 4800
BHANU earns monthly 1250 but wants 3750
JONES earns monthly 2975 but wants 8925
MANI earns monthly 1250 but wants 3750
BALU earns monthly 2850 but wants 8550
SCOTT earns monthly 3000 but wants 9000
TURNER earns monthly 1500 but wants 4500
JAMES earns monthly 950 but wants 2850

9 rows selected.
```

Question 8

Create a query to display the last name and salary for all employees. Format the salary to be 15 characters long, left-padded with the $ symbol. Label the column SALARY.

Query:

SELECT ENAME,LPAD(SAL,15,'$') AS SALARY FROM EMP;

Output:

```
SQL> SELECT ENAME,LPAD(SAL,15,'$') AS SALARY FROM EMP;

ENAME      SALARY
---------- ------------------------------------------------------------
DEEPU      $$$$$$$$$$$$800
ASHU       $$$$$$$$$$$1600
BHANU      $$$$$$$$$$$1250
JONES      $$$$$$$$$$$2975
MANI       $$$$$$$$$$$1250
BALU       $$$$$$$$$$$2850
SCOTT      $$$$$$$$$$$3000
TURNER     $$$$$$$$$$$1500
JAMES      $$$$$$$$$$$$950

9 rows selected.
```

Question 9

Display each employee's last name, hire date, and salary review date, which is the first Monday after six months of service. Label the column REVIEW. Format the dates to appear in the format similar to "Monday, the Thirty-First of July, 2000".

Query:

SELECT ENAME, HIREDATE ,TO_CHAR (NEXT_DAY (ADD_MONTHS (HIREDATE,6),'MONDAY'), 'Day, "the" DDSPTH "of" Month, YYYY') AS REVIEW FROM EMP;

Output:

```
SQL> SELECT ENAME, HIREDATE ,TO_CHAR (NEXT_DAY (ADD_MONTHS (HIREDATE,6),'MONDAY'), 'Day, "the" DDSPTH "of" Month, YYYY') AS REVIEW FR
OM EMP;

ENAME      HIREDATE  REVIEW
---------- --------- ----------------------------------------------------------------------------------------------------
DEEPU      17-DEC-80 Monday   , the TWENTY-SECOND of June     , 1981
ASHU       20-FEB-81 Monday   , the TWENTY-FOURTH of August   , 1981
BHANU      22-FEB-81 Monday   , the TWENTY-FOURTH of August   , 1981
JONES      02-APR-81 Monday   , the FIFTH of October  , 1981
MANI       28-SEP-81 Monday   , the TWENTY-NINTH of March    , 1982
BALU       01-MAY-81 Monday   , the SECOND of November , 1981
SCOTT      19-APR-87 Monday   , the TWENTY-SIXTH of October  , 1987
TURNER     08-SEP-81 Monday   , the FIFTEENTH of March    , 1982
JAMES      03-DEC-81 Monday   , the SEVENTH of June     , 1982

9 rows selected.
```

Question 10

Display the last name, hire date, and day of the week on which the employee started. Label the column DAY. Order the results by the day of the week, starting with Monday.

Query:

SELECT ENAME, HIREDATE, TO_CHAR(HIREDATE,'FMDay') AS DAY FROM EMP
ORDER BY
DECODE(TO_CHAR(HIREDATE,'FMDay'),'Monday',1,'Tuesday',2,'Wednesday',3,'Thursday',4
,'Friday',5,'Saturday',6,'Sunday',7);

Output:

```
SQL> SELECT ENAME, HIREDATE, TO_CHAR(HIREDATE,'FMDay') AS DAY FROM EMP ORDER BY DECODE(TO_CHAR(HIREDATE,'FMDay'),'Monday',1,'Tuesday'
,2,'Wednesday',3,'Thursday',4,'Friday',5,'Saturday',6,'Sunday',7);

ENAME      HIREDATE  DAY
---------- --------- ------------------------------------
MANI       28-SEP-81 Monday
TURNER     08-SEP-81 Tuesday
DEEPU      17-DEC-80 Wednesday
JONES      02-APR-81 Thursday
JAMES      03-DEC-81 Thursday
ASHU       20-FEB-81 Friday
BALU       01-MAY-81 Friday
BHANU      22-FEB-81 Sunday
SCOTT      19-APR-87 Sunday

9 rows selected.
```

Question 11

Create a query that displays the employees' last names and commission amounts. If an employee does not earn commission, show "No Commission." Label the column COMM.

Query:

SELECT ENAME, NVL(TO_CHAR(COMM),'No Commission') AS COMM FROM EMP;

Output:

```
SQL> SELECT ENAME, NVL(TO_CHAR(COMM),'No Commission') AS COMM FROM EMP;

ENAME      COMM
---------- ----------------------------------------
DEEPU      No Commission
ASHU       300
BHANU      500
JONES      No Commission
MANI       1400
BALU       No Commission
SCOTT      No Commission
TURNER     0
JAMES      No Commission

9 rows selected.
```

Question 12

Create a query that displays the first eight characters of the employees' last names and indicates the amounts of their salaries with asterisks. Each asterisk signifies a thousand dollars. Sort the data in descending order of salary. Label the column EMPLOYEES_AND_THEIR_SALARIES.

Query:

SELECT RPAD(SUBSTR(ENAME,1,8),8)||' '||RPAD('', SAL/1000,'') AS EMPLOYEES_AND_THEIR_SALARIES FROM EMP ORDER BY SAL DESC;

Output:

```
SQL> SELECT RPAD(SUBSTR(ENAME,1,8),8)||' '||RPAD('*',SAL/1000,'*') AS EMPLOYEES_AND_THEIR_SALARIES FROM EMP ORDER BY SAL DESC;

EMPLOYEES_AND_THEIR_SALARIES
--------------------------------------------------------------------------------------------------------------------------------------------
-----------------
SCOTT    ***
JONES    **
BALU     **
ASHU     *
TURNER   *
BHANU    *
MANI     *
JAMES
DEEPU

9 rows selected.
```

Question 13

Using the DECODE function, write a query that displays the grade of all employees based on the value of the column JOB_ID, using the following data: PRESIDENT-A, MANAGER-B, SALESMAN-C, CLERK-D.

Query:

SELECT ENAME, JOB,
DECODE(JOB,'PRESIDENT','A','MANAGER','B','SALESMAN','C','CLERK','D') AS GRADE FROM EMP;

Output:

```
SQL> SELECT ENAME, JOB, DECODE(JOB,'PRESIDENT','A','MANAGER','B','SALESMAN','C','CLERK','D') AS GRADE FROM EMP;

ENAME      JOB        G
---------- ---------- -
DEEPU      CLERK      D
ASHU       SALESMAN   C
BHANU      SALESMAN   C
JONES      MANAGER    B
MANI       SALESMAN   C
BALU       MANAGER    B
SCOTT      ANALYST
TURNER     SALESMAN   C
JAMES      CLERK      D

9 rows selected.
```

Question14

Rewrite the statement in the preceding exercise using the CASE syntax.

Query:

SELECT ENAME, JOB, CASE WHEN JOB='PRESIDENT' THEN 'A' WHEN JOB='MANAGER' THEN 'B' WHEN JOB='SALESMAN' THEN 'C' WHEN JOB='CLERK' THEN 'D' END AS GRADE FROM EMP;

Output:

```
SQL> SELECT ENAME, JOB, CASE WHEN JOB='PRESIDENT' THEN 'A' WHEN JOB='MANAGER' THEN 'B' WHEN JOB='SALESMAN' THEN 'C' WHEN JOB='CLERK'
THEN 'D' END AS GRADE FROM EMP;

ENAME      JOB        G
---------- ---------- -
DEEPU      CLERK      D
ASHU       SALESMAN   C
BHANU      SALESMAN   C
JONES      MANAGER    B
MANI       SALESMAN   C
BALU       MANAGER    B
SCOTT      ANALYST
TURNER     SALESMAN   C
JAMES      CLERK      D

9 rows selected.
```